HOW IT'S DESIGNED AND MADE

LET'S DIVE DEEP INTO UNDERSTANDING HOW ENGINEERS FIGURE OUT WHICH NEW FEATURES TO ADD, HOW THEY ACCOMPLISH THAT AND HOW THEY GO ABOUT MANUFACTURING THE PROCESSORS.

Mithun Mohandas | mithun@digit.in

Having learnt how CPU and GPU microarchitectures work, you must really wonder as to how they are made? How do massive technology companies such as Intel, AMD or NVIDIA figure out what direction to take with their upcoming chips? What features should they include? How should they prioritise a certain feature over the other? What if we told you that it's all decided by even bigger consortiums formed by many companies banding together. That, and lots of usage data. While that might have sounded oddly ominous as if we're living a massive corporatocracy, it really isn't the case. Well ... we're certain about the former part and not so much about the latter. Whoa! This seems to have escalated quickly, how on earth is this related to how computer chips are made? Read on.

STANDARDS AND CONSORTIUMS

What we mean by consortiums is that technology companies and scientific bodies come together to figure out which use cases require better hardware acceleration and decide on standards before going on to design their next silicon.

Take for example, AI (Artificial Intelligence), the poster child of technobabble. While the neural networks that form the very backbone of most AI frameworks have been around since the 50-60s, it's only recently that they've gained popularity. Convolutional Neural Networks or CNNs are one of the many such neural networks commonly used in building AI models. There are numerous such NNs in existence but some of them show greater promise in upcoming applications. CNNs, particularly, are excellent for AI that deals with image processing tasks. Since image processing is essential in anything that deals with images or videos such as autonomous vehicles, surveillance cameras, image search engines, video streaming services, etc. it has great potential in the near future. So companies such as Intel, AMD or NVIDIA will decide that they want to focus on accelerating CNNs since anything that can process CNNs faster is destined to sell more in the future, ergo, a CPU or GPU capable of accelerating CNNs are a guaranteed market opportunity. Hence, it would be prudent to incorporate features at the hardware level.

This was just one feature set that's obvious to all. So where do consortiums come into the picture? We hope you heard of OpenCL. It's an Open Source framework which can run across numerous hardware types such as CPUs, GPUs, FPGAs, DSPs, etc. And the only reason this happened is because the consortium called the Khronos Group decided upon the standards that make up OpenCL. Khronos is focused on authoring and accelerating playback of dynamic media. Key members of the

"We shall dominate the world!"